Tilted Cell Moments: A One-Pass Exact Newton Method for Generalized Linear Models at Scale

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July 2026

Abstract

Newton and Fisher-scoring estimation of a generalized linear model requires, at every iteration, the score vector and the information matrix of the full $N \times (p+k)$ design. When N is in the tens of millions and the design mixes k dummy-coded categorical columns with p continuous columns, the $O(N(p+k)^2)$ per-iteration cost and the memory of the design matrix itself become the binding constraints, and the standard responses—subsampling, sketching, sparse storage—either abandon exactness or retain an $O(N)$ factor on the full column dimension. This paper establishes an exact algebraic alternative. For any exponential dispersion family we prove a *tilted-moment identity*: the score and information at an arbitrary parameter value are linear combinations of three low-order moments of an exponentially tilted measure on the cells of the categorical cross-classification, computable in a single streaming pass whose cost involves only the continuous columns. The per-iteration complexity falls from $O(N(p+k)^2)$ to $O(N(p^2+p+1)) + O(J(k^2+kp))$ for J cells, no design matrix is ever formed, and the resulting iterates coincide with dense Newton to floating-point reordering (10^{-13}–10^{-15} in experiments across Poisson, logistic, and gamma models): the method is exact, not an approximate-gradient scheme. We derive the identity, characterize precisely why no fixed cell summary can support the likelihood function itself (a Laplace-transform uniqueness argument covering the Poisson and Bernoulli families), give the streaming algorithm with its numerical-stability envelope (log-domain kernels, backtracking, rank-deficiency and separation handling) and exact cell-accumulated sandwich and cluster-robust covariance estimators, and benchmark a reference implementation against dense Newton, sparse-design IRLS, and production solvers: one to two orders of magnitude faster and 12–50× lighter in memory than `statsmodels`, faster than `glum` and sparse IRLS, with $O(N)$ time and memory verified to 5×10^7 observations and accuracy preserved at

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information-matrix condition numbers up to 10^{17}. As a stress test, the engine composes with high-dimensional fixed-effect absorption—profile updates of absorbed factors alternated with tilted-moment Newton steps—for which we supply a unified global convergence proof; on public data with 3,692 absorbed fixed effects the composition reproduces `pyfixest` to 10^{-9} at lower cost.

Keywords: generalized linear models; Newton–Raphson; Fisher scoring; exponential tilting; streaming algorithms; computational complexity; high-dimensional fixed effects.

1 Introduction

Maximum-likelihood estimation of a generalized linear model (Nelder and Wedderburn, 1972; McCullagh and Nelder, 1989) proceeds by Newton–Raphson or Fisher scoring: at each iteration, form the score vector and the information matrix of the current iterate and solve a q-dimensional linear system, $q = p + k$ the number of coefficients. With the design held as an $N \times q$ matrix, each iteration costs $O(Nq^2)$ floating-point operations for the weighted Gram product, and the design itself occupies Nq doubles. Neither term is problematic at textbook scale. Both become binding when N reaches 10^7–10^8 and the design has the composition typical of administrative, clinical, and operational data: a small number p of continuous covariates alongside a rich categorical apparatus—strata, batches, regions, periods, and their interactions—whose dummy coding contributes $k \gg p$ columns. A design of 5×10^7 rows and 500 dummy columns occupies 200 GB before the first iteration; the Gram product touches all of it, every iteration. This is the bottleneck the present paper removes.

The removal is algebraic, not approximate. Let the categorical covariates partition the observations into the J cells of their cross-classification, and let the k dummy-coded columns be constant within cells while the p continuous columns vary freely. Two classical facts frame the problem. When $p = 0$, the exponential-family log likelihood collapses onto within-cell aggregates, and estimation on the J cells reproduces the microdata maximum-likelihood estimator (MLE) exactly—the standard device of grouped Poisson rate modeling (Frome, 1983; Clayton and Hills, 1993; Breslow and Day, 1987) and, for binary data, of grouped binomial fitting. And the collapse is fragile: we show (Proposition 1) that for $p \geq 1$ *no* fixed-dimensional cell summary can support the likelihood *function*, because evaluating the within-cell cumulant sum at all parameter values determines the Laplace transform of the within-cell empirical measure of the continuous covariates—and Laplace transforms determine measures. The argument is not Poisson folklore: we prove it for the Poisson and Bernoulli families by one uniqueness argument, so the impossibility is a property of the exponential family, not of a particular application.

The contribution of the paper is the observation that estimation does not require the likelihood function; it requires the gradient and curvature *at the current iterate*, and these admit an exact finite representation that the function itself does not. Theorem 1—the tilted-moment identity—states that for any exponential dispersion family with canonical link, the score vector and information matrix at an arbitrary parameter value are linear combinations of three low-order moments ($M_j^{(0)}$, $\mathbf{m}_j$, $\mathbf{Q}$) of an *exponentially tilted* measure on each cell. For the Poisson family the tilt is an Esscher transform of the within-cell exposure measure, free of the cell-level coefficients; for the Bernoulli–logit family it is the variance-weighted measure that concentrates on observations near the decision boundary; the identity is the same in both cases, and Fisher scoring under non-canonical links has the same structure (Proposition 2), with the gamma–log family as the limiting case in which the tilt weights are identically one and the moments are constant across iterations. The tilt is the correct *dynamic* object: a fixed collapse fails because it attempts likelihood exactness, which Proposition 1 forbids; the tilted collapse succeeds because it needs only score exactness, iterate by iterate.

The identity converts directly into complexity, which is the paper's scaling proposition (Corollary 1): one Newton or Fisher-scoring iteration costs a single streaming $O(N(p^2+p+1))$ pass over the microdata—the continuous columns only—plus $O(J(k^2+kp))$ cell-level assembly and one $O((p+k)^3)$ solve, against $O(N(p+k)^2)$ for the full design. The saving grows quadratically in the categorical richness k; the memory footprint contains no Nk term at all, since no design matrix is ever formed; and, because the identity is exact rather than a linearization, the iterates coincide with dense Newton up to floating-point reordering, which Section 5.1 verifies at 10^{-13}–10^{-15}, iterate by iterate, for Poisson, logistic, and gamma models. The distinction matters for how the method should be classified: it is not a stochastic-gradient, subsampled, or sketched solver with an accuracy–cost dial, but an exact reorganization of Newton's arithmetic whose output *is* the MLE with its full inferential apparatus—observed information, sandwich, and cluster-robust covariance matrices, all accumulated cellwise in the same pass (Huber, 1967; White, 1980; Liang and Zeger, 1986).

The second half of the paper develops the identity into a solver and measures it. Section 4 gives the streaming algorithm and its numerical envelope: log-domain likelihood kernels, backtracking line search, rank-deficiency and separation handling (Wedderburn, 1976; Albert and Anderson, 1984), exact cell-accumulated covariance estimators, memory layout, and parallel structure. Section 5 contains the experiments, organized around three comparisons: dense Newton as the ground-truth baseline for exactness; sparse-design IRLS—the strongest general-purpose competitor, since dummy columns are the textbook sparse design—for the like-for-like cost comparison; and the production solvers `statsmodels` (Seabold and Perktold, 2010) and `glum` (Balzer et al., 2021) for real-world viability, together with $O(N)$ scaling to 5×10^7 observations and a numerical-stability battery spanning condition numbers to 10^{17}. Section 6 exercises the solver on public data (a logistic on-time-performance model for the 336,776 flights of `nycflights13`, verified against `statsmodels` to 10^{-13}) and on a 5×10^6-observation Poisson benchmark.

One structured application receives separate treatment because it is the natural stress test of the engine: high-dimensional fixed-effect (HDFE) absorption, where one or more factors—firms, aircraft, region–period interactions (Abowd et al., 1999)—have too many levels even for cell-level treatment and are concentrated out by profile updates alternated with tilted-moment Newton steps (Section 4.5). The composition inherits the engine's economics: the absorbed factors never enter any design either. Convergence of the linear inner demeaning loop is classical—the method of alternating projections of von Neumann and Halperin (1962), brought into fixed-effects estimation by Gaure (2013)—and the combined nonlinear GLM cycle has long been validated by simulation (Guimarães and Portugal, 2010; Bergé, 2018; Correia et al., 2020); what has been missing is a single formal theorem for the combined inner–outer iteration, which we state as Theorem 2 (proof in the supplement) and verify against `pyfixest` on real data with 3,692 absorbed effects to 10^{-9}. The supplement develops the associated rate and asymptotic theory; in the main text the absorption path is strictly an application of the tilting engine, not a second subject.

Relation to existing computational methods. The method occupies a specific point in the landscape of large-N GLM computation, and its novelty is best stated by contrast. *Approximate data-reduction methods*—optimal subsampling for logistic regression (Wang et al., 2018), information-based subdata selection (Wang et al., 2019), leverage-score sampling (Ma et al., 2015), local case-control sampling for rare events (Fithian and Hastie, 2014), and randomized sketching (Drineas and Mahoney, 2016)—trade statistical accuracy for cost along a tunable frontier; the tilted pass is exact and deterministic, so where a cell structure exists it dominates this frontier pointwise rather than moving along it. *Collapsed sufficient statistics* for grouped rate and binomial models (Frome, 1983; Clayton and Hills, 1993) are the special case $p = 0$; the tilted moments extend the exact collapse to $p \geq 1$, which is precisely the case Proposition 1 shows a fixed collapse cannot reach. *Matrix-free and Hessian-free Newton methods* (Pearlmutter, 1994; Martens, 2010; Nocedal and Wright, 2006) avoid storing the Hessian by supplying Hessian–vector products to an iterative solver; they still stream the full design through every product and solve the q-dimensional system approximately, whereas the tilted representation forms the *exact* $(p + k)$-dimensional system in closed form from cell moments and solves it directly. *Grouped or stochastic second-order methods* (Byrd et al., 2011) subsample curvature; the tilted step is exact. *Automatic differentiation* (Baydin et al., 2018) mechanizes the same derivatives at $O(N(p + k)^2)$; the contribution here is the analytic identity that makes the derivative a cell-level object. *Sparse-design and bounded-memory solvers* (`glum`, Balzer et al., 2021; `biglm`, Lumley, 2011) store the dummy block sparsely or stream the design in chunks; Section 5.2 shows the sparse route is the strongest general competitor, and that the tilted pass wins because it never materializes a design at all and its cost grows only in the continuous dimension. *Distributed and communication-avoiding optimization* (Zhang et al., 2013; Jaggi et al., 2014) splits one fit across machines, an orthogonal axis: the tilted accumulators are additive across data partitions, so the pass distributes trivially per shard. Finally, the *HDFE stack* (Guimarães and Portugal, 2010; Gaure, 2013; Correia, 2016; Correia et al., 2020; Bergé, 2018; Stammann, 2018) removes dummy factors by iterative demeaning but streams the full non-absorbed design through every IRLS step; the two devices compose (Section 4.5), with the tilted pass handling the cell-structured block that HDFE tools treat as ordinary columns. In summary: the contribution is neither "a sufficient statistic exists" nor "regroup the sums," but the exact, iterate-dependent tilted-moment identity (Theorem 1) that turns the score and the full information matrix of a mixed categorical–continuous GLM into a one-pass cell computation, together with its complexity, memory, and stability consequences.

All numerical results regenerate from a self-contained open-source Python/NumPy reference implementation released with the manuscript; a companion Stata command (`stpois`), distributed in the same repository, implements the Poisson event-history case for applied use.

2 Exponential family models and the failure of ordinary collapse

2.1 Model and notation

Observations $i = 1, \ldots, N$ carry a response y_i, a vector $\mathbf{x}_i \in \mathbb{R}^p$ of *individual-level* covariates (those involving at least one continuous variable, including continuous–categorical interaction columns), a vector $\mathbf{w}_{j(i)} \in \mathbb{R}^k$ of *cell-level* covariates, and a known offset o_i. The map $j : \{1, \ldots, N\} \to \{1, \ldots, J\}$ assigns each observation to the cell of the cross-classification of the categorical covariates; $\mathbf{w}_j$ is constant within cell j by construction. Conditional on the covariates, responses are independent draws from an exponential dispersion family

$$f(y_i \mid \theta_i, \phi) = \exp\left\{ \frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y_i, \phi) \right\}, \tag{1}$$

with $b : \Theta \to \mathbb{R}$ strictly convex and three times continuously differentiable on the open canonical parameter space $\Theta \subseteq \mathbb{R}$, and dispersion $\phi > 0$ (Nelder and Wedderburn, 1972; Jørgensen, 1987; McCullagh and Nelder, 1989). The mean and variance are $\mu_i = b'(\theta_i)$ and $\mathrm{Var}(y_i) = \phi\, b''(\theta_i)$. We model the linear predictor

$$\eta_i = \boldsymbol{\gamma}'\mathbf{x}_i + \boldsymbol{\delta}'\mathbf{w}_{j(i)} + o_i, \qquad \boldsymbol{\theta} = (\boldsymbol{\gamma}', \boldsymbol{\delta}')' \in \mathbb{R}^q, \quad q = p + k, \tag{2}$$

and, until Section 3.2, take the link to be canonical, $\theta_i = \eta_i$. Two members of the family serve as running examples throughout, on equal footing. *Poisson*: $b(\theta) = e^\theta$, $\phi = 1$, y_i a count, and $o_i = \log t_i$ a log exposure; with episode-split survival data this is the piecewise-exponential event-history likelihood (Holford, 1980; Aitkin and Clayton, 1980; Andersen et al., 1993). *Bernoulli–logit*: $b(\theta) = \log(1 + e^\theta)$, $\phi = 1$, $y_i \in \{0, 1\}$—the workhorse for binary outcomes at scale. The gamma family with log link enters in Section 3.2 as the canonical illustration of the non-canonical case. Up to terms not involving $\boldsymbol{\theta}$, the log likelihood is

$$\ell(\boldsymbol{\theta}) = \frac{1}{\phi} \sum_{i=1}^{N} \{ y_i \eta_i - b(\eta_i) \}. \tag{3}$$

Because (3) is a sum of concave functions of the linear predictor, ℓ is concave on $\mathbb{R}^q$, strictly so on the quotient of $\mathbb{R}^q$ by the null space of the stacked design $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$. We write $\mathbf{U}(\boldsymbol{\theta}) = \nabla \ell$ for the score, $\mathbf{H}(\boldsymbol{\theta}) = -\nabla^2 \ell$ for the negative Hessian, and note that for canonical links $\mathbf{H}$ equals the Fisher information $\mathbf{F}(\boldsymbol{\theta}) = \mathbb{E}[-\nabla^2 \ell]$, so Newton–Raphson and Fisher scoring coincide.

2.2 Cell collapse and its failure

Suppose first that $p = 0$: every covariate is categorical, so $\eta_i = \boldsymbol{\delta}'\mathbf{w}_j + o_i$ varies over i within a cell only through the offset. For the Poisson family with $o_i = \log t_i$, (3) becomes $\ell(\boldsymbol{\delta}) = \sum_j \{D_j\, \boldsymbol{\delta}'\mathbf{w}_j - T_j e^{\boldsymbol{\delta}'\mathbf{w}_j}\} + \text{const}$, with $D_j = \sum_{i\in j} y_i$ and $T_j = \sum_{i\in j} t_i$: the collapsed cell data (D_j, T_j) carry all the information. For the Bernoulli–logit family with constant offsets the same collapse holds with (D_j, n_j), the cell event count and cell size—grouped binomial fitting. In both cases estimation on the J cells is exact, at $O(Jq^2)$ per iteration. The following proposition records why no analogous finite collapse survives the introduction of a continuous covariate, and in what precise sense; the argument is a property of the family's cumulant function, and we give it in one form that covers both running examples. Define the within-cell *weighted covariate measure* $\nu_j = \sum_{i\in j} e^{o_i}\delta_{\mathbf{x}_i}$ on $\mathbb{R}^p$ (for Poisson, the exposure measure $\nu_j = \sum_{i\in j} t_i\delta_{\mathbf{x}_i}$; for Bernoulli without offsets, the empirical measure of the continuous covariates), a finite discrete measure recording the joint configuration of offsets and continuous covariates in cell j, and write $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) = \int e^{\boldsymbol{\gamma}'\mathbf{x}}\,\nu_j(\,\mathrm{d}\mathbf{x})$ for its Laplace transform.

Proposition 1 (Failure of fixed collapse)**.** *Let the cumulant b be real-analytic on $\mathbb{R}$ with $b(u) = e^u\,(1 + o(1))$ as $u \to -\infty$—as holds for the Poisson family, where $b(u) = e^u$ exactly, and the Bernoulli family, where $b(u) = \log(1 + e^u)$. (i) If $(\mathbf{x}_i, o_i) = (\bar{\mathbf{x}}_j, \bar{o}_j)$ for all $i \in j$ and all j, the log likelihood (3) depends on the data only through $\{(D_j, n_j, \bar{\mathbf{x}}_j, \bar{o}_j, \sum_{i\in j} y_i\mathbf{x}_i)\}_{j\le J}$, and the MLE on the collapsed cells equals the microdata MLE; for Poisson the offset need not be constant, and (D_j, T_j) suffices as above. (ii) In general, $\ell(\boldsymbol{\gamma}, \boldsymbol{\delta})$ depends on the data through $\{S_j, D_j, G_j\}_{j\le J}$, where $S_j = \sum_{i\in j} y_i\mathbf{x}_i$ and $G_j(s, \boldsymbol{\gamma}) = \sum_{i\in j} b(s + \boldsymbol{\gamma}'\mathbf{x}_i + o_i)$ is the within-cell cumulant functional; and any statistic from which the likelihood function can be evaluated for $(\boldsymbol{\gamma}, \boldsymbol{\delta})$ in an open set determines G_j on an open set, hence—by real-analyticity in s and the boundary expansion $e^{-s}G_j(s, \boldsymbol{\gamma}) \to \mathcal{L}_{\nu_j}(\boldsymbol{\gamma})$ as $s \to -\infty$—determines the Laplace transform of ν_j, and therefore ν_j itself. The support size of ν_j, and with it the dimension of any such statistic, grows with the number of distinct covariate values in the cell: no fixed-dimensional cell summary can support the likelihood function. (iii) By strict convexity of the exponential (Jensen), $\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) \ge \nu_j(\mathbb{R}^p)\, e^{\boldsymbol{\gamma}'\bar{\mathbf{x}}_j}$ with equality along a direction of $\boldsymbol{\gamma}$ only if $\boldsymbol{\gamma}'\mathbf{x}$ is ν_j-a.s. constant along it: replacing microdata by cell means strictly understates the within-cell mass of the cumulant term and is not a reparametrization of the same likelihood—the same convexity mechanism that biases log-linearized estimation relative to Poisson pseudo-maximum-likelihood (Santos Silva and Tenreyro, 2006).*

Part (ii) is the formal statement of the practical dictum that "continuous covariates break collapsibility," and it holds uniformly across the family: the fixed-dimensional sufficient statistic $\sum_i y_i\mathbf{a}_i$ for the canonical parameter still exists, but *computing the likelihood function*—and hence running any optimizer that evaluates it—requires functionals of the data whose dimension is of order N, not J. The next section shows that the required functionals nevertheless have a special structure at any *single* parameter value: they are three low-order moments of a tilted measure, computable in one linear pass. Estimation needs only this pointwise object.

3 Tilted cell moments

3.1 Exponentially tilted cell moments

Definition 1 (Tilted within-cell measures)**.** *Fix $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\delta})$ and let $\omega_i = b''(\eta_i) > 0$ be the information weight of observation i. The* information-tilted measure *on cell j is the probability measure*

$$P_{j,\boldsymbol{\theta}}(\{i\}) = \frac{\omega_i}{\sum_{i' \in j} \omega_{i'}}, \qquad i \in j.$$

For the Poisson family, $\omega_i = \mu_i = t_i e^{\boldsymbol{\delta}'\mathbf{w}_j + \boldsymbol{\gamma}'\mathbf{x}_i}$, the factor $e^{\boldsymbol{\delta}'\mathbf{w}_j}$ cancels, and $P_{j,\boldsymbol{\theta}}$ reduces to the Esscher transform *(exponential tilt) of the normalized exposure measure ν_j/T_j:*

$$P_{j,\boldsymbol{\gamma}}(\,\mathrm{d}\mathbf{x}) = \frac{e^{\boldsymbol{\gamma}'\mathbf{x}}\,\nu_j(\,\mathrm{d}\mathbf{x})}{\mathcal{L}_{\nu_j}(\boldsymbol{\gamma})}, \tag{4}$$

free of $\boldsymbol{\delta}$. Its moments are derivatives of the within-cell cumulant generating function: with $K_j(\boldsymbol{\gamma}) = \log \mathcal{L}_{\nu_j}(\boldsymbol{\gamma})$, $\mathbb{E}_{P_{j,\boldsymbol{\gamma}}}[\mathbf{x}] = \nabla K_j(\boldsymbol{\gamma})$ and $\mathrm{Var}_{P_{j,\boldsymbol{\gamma}}}(\mathbf{x}) = \nabla^2 K_j(\boldsymbol{\gamma})$.

The tilt (4) is precisely the change of measure that underlies exponential-family conjugation and saddlepoint approximation (Barndorff-Nielsen, 1978; Daniels, 1954; Jensen, 1995) and exponential-tilting estimators in the empirical-likelihood family (Efron, 1981; Kitamura and Stutzer, 1997; Owen, 2001; Schennach, 2007); here it appears not as an approximation device but as an exact algebraic identity for the score and information, as follows. Write, for each cell j,

$$M_j^{(0)} = \sum_{i \in j} \omega_i, \qquad \mathbf{m}_j = \sum_{i \in j} \omega_i \mathbf{x}_i, \qquad \mathbf{Q} = \sum_{i=1}^{N} \omega_i \mathbf{x}_i \mathbf{x}_i', \tag{5}$$

and let $B_j = \sum_{i \in j} b'(\eta_i)$ and $R_j = \sum_{i \in j} (y_i - b'(\eta_i))$ denote the cell sums of means and of raw residuals.

Theorem 1 (Exact tilted-moment representation)**.** *For the model (1)–(2) with canonical link, at every $\boldsymbol{\theta} \in \mathbb{R}^q$:*

(i) The score is

$$\phi\,\mathbf{U}(\boldsymbol{\theta}) = \begin{pmatrix} \sum_i y_i \mathbf{x}_i - \sum_j \mathbf{m}_j^{b'} \\ \sum_j \mathbf{w}_j R_j \end{pmatrix}, \qquad \mathbf{m}_j^{b'} = \sum_{i \in j} b'(\eta_i)\,\mathbf{x}_i = B_j\,\mathbb{E}_{P^{b'}_{j,\boldsymbol{\theta}}}[\mathbf{x}],$$

where $P^{b'}_{j,\boldsymbol\theta}$ is the tilt of Definition 1 with b' in place of b''.

(ii) The information matrix is

$$\phi\,\mathbf{F}(\boldsymbol\theta) = \begin{pmatrix} \mathbf{Q} & \sum_j \mathbf{m}_j\mathbf{w}_j' \\ \sum_j \mathbf{w}_j\mathbf{m}_j' & \sum_j M_j^{(0)}\mathbf{w}_j\mathbf{w}_j' \end{pmatrix}, \qquad \mathbf{m}_j = M_j^{(0)}\,\mathbb{E}_{P_{j,\boldsymbol\theta}}[\mathbf{x}], \quad \mathbf{Q} = \sum_j M_j^{(0)}\,\mathbb{E}_{P_{j,\boldsymbol\theta}}[\mathbf{x}\mathbf{x}'].$$

(iii) For the Poisson family, $b' = b''$, the two tilts coincide with the $\boldsymbol\delta$-free Esscher transform (4), and $B_j = M_j^{(0)} = e^{\boldsymbol\delta'\mathbf{w}_j}\mathcal{L}_{\nu_j}(\boldsymbol\gamma)$, $\mathbf{m}_j = M_j^{(0)}\nabla K_j(\boldsymbol\gamma)$, and the $\boldsymbol\gamma$-block of $\mathbf{F}$ is $\sum_j M_j^{(0)}\{\nabla^2K_j(\boldsymbol\gamma) + \nabla K_j\nabla K_j'\}$. For the Bernoulli–logit family, $b''(\eta_i) = \mu_i(1-\mu_i)$: the information tilt weights each within-cell observation by its Bernoulli variance, concentrating $P_{j,\boldsymbol\theta}$ on observations near the decision boundary; the tilt depends on $\boldsymbol\delta$ through μ_i, so the moments (5) are re-accumulated at each iterate—which is exactly what the single pass does.

(iv) The Newton step $\boldsymbol\theta^+ = \boldsymbol\theta + \mathbf{F}(\boldsymbol\theta)^{-1}\mathbf{U}(\boldsymbol\theta)$ computed from the quantities in (i)–(ii) is identical to the Newton step computed from the full $N\times q$ design.

Claim (iv) is immediate once (i)–(ii) are established—the two computations produce the same matrices—but it is the operative point: the decomposition involves no linearization, discretization, or stochastic approximation, so the accelerated algorithm is not a fast approximate estimator but a fast exact organization of the arithmetic of the MLE. Section 5.1 exhibits agreement of the full iterate *sequences* at the 10^{-13} level, which is the accumulation of floating-point reordering only.

Statistically, the representation has content beyond bookkeeping. The $\boldsymbol\delta$-score in (i) compares each cell's observed event mass with its expected mass under the current tilt; the cross-information in (ii) is the tilted covariance between cell-level and individual-level design; and the $\boldsymbol\gamma$-block in (iii) decomposes into a within-cell tilted covariance ∇^2K_j plus a between-cell term. In particular the identification of $\boldsymbol\gamma$ rests exactly on $\sum_j M_j^{(0)}\nabla^2K_j(\boldsymbol\gamma) \succ \mathbf{0}$: continuous covariates are identified through their *within-cell, tilt-weighted* variation—a fact that the block structure displays, and that drives the Schur-complement profile Hessians used throughout the companion theory in the online supplement. (When a continuous covariate is constant within cells—a route-level distance, say—it remains identified through between-cell variation as long as the cell-level design $\mathbf{w}_j$ is not saturated; Section 6.1 exercises this case.)

Corollary 1 (Scaling)**.** *One Newton iteration computed via Theorem 1 costs*

$$\underbrace{O\Big(N(p^2+p+1)\Big)}_{\textit{microdata pass}} + \underbrace{O\Big(J(k^2+kp)\Big)}_{\textit{cell assembly}} + \underbrace{O\Big((p+k)^3\Big)}_{\textit{solve}}$$

floating-point operations and one $O(N)$ memory sweep, versus $O(N(p+k)^2) + O((p+k)^3)$

for the full-design iteration. When $N \gg J(p+k)$ the leading-order saving is the factor $(p+k)^2/(p^2+p+1)$, which grows quadratically in the richness k of the categorical structure; when additionally $p = 0$ the microdata pass degenerates to the classical cell collapse and the entire iteration runs at $O(Jk^2)$.

Two remarks on the constant factors. First, the microdata pass is a fused stream: one evaluation of $b'(\eta_i)$, $b''(\eta_i)$ per observation, $p+1$ multiply–accumulate reductions into cell bins, and a rank-one update of the $p \times p$ Gram matrix; on modern hardware it is memory-bandwidth-bound, so measured speedups (Section 5.1) are large but smaller than the FLOP ratio. Second, the accumulations are plain segmented sums—`bincount` operations—so they parallelize and stream trivially; no $N \times q$ design matrix is ever formed, which brings a problem whose dense design exceeds working memory (Section 6.2) within a laptop's footprint.

3.2 Non-canonical links

So far the link has been canonical, $\theta_i = \eta_i$, where the Newton weight $b''(\eta_i)$ and the tilt coincide. The construction does not depend on this. For a general link g with $\mu_i = g^{-1}(\eta_i)$ the maximum-likelihood score for a linear exponential family is $\sum_i (y_i - \mu_i)\, \frac{\partial \mu_i / \partial \eta_i}{V(\mu_i)}\, \mathbf{a}_i$ and the expected (Fisher) information is $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$, with the standard IRLS weight and working residual

$$\omega_i = \frac{1}{V(\mu_i)\, g'(\mu_i)^2}, \qquad r_i = (y_i - \mu_i)\, g'(\mu_i)\, \omega_i \tag{6}$$

(McCullagh and Nelder, 1989, Sec. 2.5)—the quasi-likelihood weighting of Wedderburn (1974), whose IRLS/Gauss–Newton correspondence is developed by Green (1984)—where V is the variance function and $g'(\mu) = \partial \eta / \partial \mu$. Comparing with Theorem 1, the only change is that the pair $(b'(\eta_i), b''(\eta_i)) = (\mu_i, \omega_i^{\text{can}})$ of the canonical case is replaced by (μ_i, ω_i) and the raw residual $y_i - \mu_i$ by r_i: the tilt is now weighted by the Fisher weight (6), and everything downstream—the cell sums, the Gram matrix, the block assembly—is unchanged.

Proposition 2 (Tilted Fisher scoring)**.** *With ω_i and r_i as in (6), the expected-information scoring step admits the representation of Theorem 1, with $M_j^{(0)}, \mathbf{m}_j, \mathbf{Q}$ built from ω_i and $R_j = \sum_{i \in j} r_i$; the fixed points of the iteration are exactly the roots of the ML score equations, and Corollary 1 applies verbatim.*

Two points of intuition and scope. First, whether the moments must be *re-accumulated* each iteration depends on how ω_i depends on η_i. For canonical links $\omega_i = b''(\eta_i)$ moves with the iterate, so a fresh pass is needed—which is the pass the algorithm takes anyway. The gamma family with log link is the striking exception: $V(\mu) = \mu^2$ and $g'(\mu) = 1/\mu$ give $\omega_i \equiv 1$, so the Fisher weight is constant, the "tilted" moments are ordinary exposure-weighted cell moments *fixed across iterations*, and after one pass each subsequent iteration costs $O(Np) + O(Jk^2)$—the acceleration is at its most extreme precisely where the link is non-canonical. Second, the representation needs only that $\omega_i > 0$ and V, g be smooth on the range of the fitted means, so it covers the standard non-canonical GLMs—gamma or inverse-Gaussian with log link, binomial with probit or complementary log-log, Poisson with identity or square-root link—as expected-information scoring. It does not by itself guarantee that scoring *converges globally* for these links; that requires the log-concavity condition made precise for the absorbed-factor problem in Supplement Theorem S1, which holds for gamma–log and probit but not for every link. Observed information (true Newton) under a non-canonical link adds a term in b'''/g'' that is equally cell-decomposable but that we do not need, since Fisher scoring already delivers the MLE at the same per-iteration cost.

3.3 Families in scope

Three instances are carried through the experiments, chosen to span the structural cases rather than to catalogue applications. *Poisson with log link* (canonical; $\boldsymbol{\delta}$-free Esscher tilt (4), moments computable from within-cell cumulant generating functions) and *Bernoulli with logit link* (canonical; $\boldsymbol{\delta}$-dependent variance tilt, moments re-accumulated per iterate) are the two running examples and receive equal treatment throughout. *Gamma with log link* (non-canonical, Proposition 2) is the boundary case of the Fisher weight: $\omega_i \equiv 1$, so the "tilted" moments are ordinary cell moments, fixed across iterations, and after the first pass each iteration costs $O(Np) + O(Jk^2)$. Together the three exhibit the full range of tilt behavior—exactly exponential, iterate-dependent, and constant.

4 Algorithms and implementation

4.1 The streaming pass

Algorithm 1 states the method operationally. Its preprocessing is one factorization of the categorical covariates into cell identifiers $j(i)$ and one construction of the $J \times k$ cell-level design $\mathbf{W}$; both are $O(N)$ given a sort or hash of the categorical columns, and both are shared across models on the same cells.

By Theorem 1(iv) the iterate sequence is identical to full-design Newton from the same start; every design decision below concerns constants, not correctness.

4.2 Numerical safeguards

Three matter in practice. (i) *Monotonicity*: the backtracking line search on the log likelihood guarantees ascent and terminates, since the Newton direction of a positive-definite system is an ascent direction; in well-conditioned problems the full step is accepted after the first iterations and convergence is quadratic. (ii) *Log-domain evaluation*: likelihood and mean

Algorithm 1 Tilted-moment Newton / Fisher scoring

1: Precompute cell ids $j(i)$, cell design $\mathbf{W} \in \mathbb{R}^{J\times k}$, continuous columns $\mathbf{X} \in \mathbb{R}^{N\times p}$, offsets o_i; initialize $\boldsymbol{\theta}_0 = (\boldsymbol{\gamma}_0, \boldsymbol{\delta}_0)$.
2: **repeat**
3: *Stream over $i = 1, \dots, N$:* compute $\eta_i = \boldsymbol{\gamma}'\mathbf{x}_i + (\mathbf{W}\boldsymbol{\delta})_{j(i)} + o_i$, the weight ω_i, and the residual r_i; accumulate $M^{(0)}_{j(i)} \mathrel{+}= \omega_i$; $\mathbf{m}_{j(i)} \mathrel{+}= \omega_i \mathbf{x}_i$; $R_{j(i)} \mathrel{+}= r_i$; $\mathbf{U}_\gamma \mathrel{+}= r_i \mathbf{x}_i$; $\mathbf{Q} \mathrel{+}= \omega_i \mathbf{x}_i \mathbf{x}_i'$.
4: *Cell assembly:* $\mathbf{U}_\delta = \mathbf{W}'\mathbf{R}$; $\mathbf{F}_{\gamma\delta} = \mathbf{m}'\mathbf{W}$; $\mathbf{F}_{\delta\delta} = \mathbf{W}' \operatorname{diag}(M^{(0)})\mathbf{W}$; $\mathbf{F}_{\gamma\gamma} = \mathbf{Q}$.
5: Solve the $(p+k)$-dimensional system $\mathbf{F}\Delta = \mathbf{U}$; update $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \lambda\Delta$ with backtracking $\lambda \in \{1, \frac{1}{2}, \dots\}$ on the log likelihood.
6: **until** $\max_\ell |\Delta\theta_\ell|/(1 + |\theta_\ell|) < \tau$

evaluations use `log1p` / `logaddexp`-style kernels so that extreme linear predictors (produced, e.g., en route to a separated fit) yield $-\infty$ log likelihoods rather than overflow. (iii) *Rank deficiency and separation*: existence and uniqueness of the GLM MLE can fail (Wedderburn, 1976), classically under complete or quasi-complete separation in the binary case (Albert and Anderson, 1984) and under analogous boundary configurations for Poisson (Santos Silva and Tenreyro, 2010, 2011; Correia et al., 2019). If the assembled $\mathbf{F}$ is singular—collinear cell-level columns, or vanishing weights under quasi-separation—the solve falls back to a minimum-norm (pseudoinverse) direction and the iteration runs to its cap with the non-convergence flagged, never silently; separation is a property of the data, not of the algorithm. Section S3 of the supplement develops the bias-reducing adjusted-score variant of Firth (1993)—the standard finite-maximizer remedy (Heinze and Schemper, 2002; Kosmidis and Firth, 2009)—computed inside the same streaming pass at the same per-iteration order.

4.3 Memory layout and parallelism

The gap between measured and FLOP ratios merits a constructive comment, since closing it is an implementation exercise with a known playbook. The tilted pass streams $O(Np)$ numbers and performs $O(Np^2)$ flops, so for small p its arithmetic intensity is low and the memory bus, not the ALU, sets the speed; the dense baseline, by contrast, is a BLAS-3 Gram update running near peak. Three standard remedies raise the tilted pass toward bandwidth saturation and beyond: (i) *cache locality*: storing the microdata sorted by cell converts the binned accumulations into contiguous segmented sums, so each cell's partial moments stay in registers/L1 and the pass touches each cache line once, the streaming analogue of the classical one-pass moment updates of Welford (1962) (cf. Correia, 2016); (ii) *fusion*: the weight evaluation, residual, binned sums, and Gram update share one loop, so the data are read once per iteration rather than once per moment; and (iii) *parallelism*: cells are independent, so the pass shards trivially across threads with per-thread partial moments and a Jk-sized reduction, with no atomics on the hot path—the same structure GPU segmented-reduction primitives expect. The single-threaded

NumPy reference implementation uses none of (i)–(iii) beyond what `bincount` provides, so the reported speedups should be read as a floor; the FLOP ratio governs how the comparison scales with (p, k), and the bandwidth analysis governs the attainable constant.

The *memory* footprint is where the reorganization is most decisive, and it is easy to state exactly. The tilted pass holds the loaded columns, $O(N(p + 1))$ doubles, plus the accumulators: J cell scalars, a $J \times p$ moment array, and the $(p + k) \times (p + k)$ information—$O(N(p+1)) + O(J(k+p) + (p+k)^2)$ in total, with no term in Nk. A dense design is $N(p+k)$ doubles, and a general-purpose IRLS adds working copies on top; the measured peak resident sets bear this out (Tables 2 and 5): ~ 80 MB for the tilted fit against ~ 1 GB dense and $\sim$4 GB for `statsmodels` at $N = 2 \times 10^5$, widening to 3.8 GB versus a 200 GB dense design at $N = 5 \times 10^7$. Because peak memory, not FLOPs, is what makes a register-scale fit feasible or not, this is the practically load-bearing comparison. Preprocessing—the one-time sort or hash that builds the cell identifiers and, for absorbed factors, the group indices—is $O(N \log N)$ or $O(N)$ and is shared across all models fitted on the same cells (a sweep of specifications, a bootstrap, a cross-validation), so it is reported separately from the per-iteration cost throughout and is amortized whenever more than one model is fitted on a design.

Reproducibility of the timings. All wall-clock and peak-memory figures are measured on a single core of a commodity x86-64 workstation (AVX2, 32 KiB L1 / 1 MiB L2 per core, shared L3), with each competitor fitted in a fresh process so that peak resident set isolates its footprint; times are medians over repeated fits. The reference implementation is pure NumPy/SciPy on the system BLAS, single-threaded (BLAS threads pinned to one) so that the comparison reflects algorithmic organization rather than parallel speedup, which Section 4.3 argues the tilted pass admits but does not yet use. The dense-side factorizations and the pseudoinverse fallback follow standard practice (Golub and Van Loan, 2013). The exact hardware affects the constants, not the ratios or the $O(\cdot)$ trends, which are the paper's claims; every figure regenerates from the scripts in the code supplement.

4.4 Robust and clustered variance estimation

All variance estimators used in practice share the cell structure. The observed/expected information is $\mathbf{F}$ itself (Theorem 1). The Huber–White sandwich estimator (Huber, 1967; White, 1980, 1982)—the default under conditional-mean correctness without full distributional trust (Gouriéroux et al., 1984)—needs $\sum_i \hat{u}_i^2 \mathbf{a}_i \mathbf{a}_i'$ with $\hat{u}_i = y_i - \hat{\mu}_i$; its blocks accumulate cellwise exactly like (5) with ω_i replaced by $\hat{u}_i^2$. The cluster-robust estimator (Liang and Zeger, 1986; Cameron et al., 2011) sums scores within clusters before the outer product; when clusters are unions of cells (the common design: cluster on region, cells within region) the accumulation is again one pass. These identities are what make *inference*, not just point estimation, run at cell-collapsed cost; they are implemented in the reference code and validated against dense computation in the test suite.

4.5 A stress test: absorbed fixed-effect factors

The natural stress test of the tilting engine is the setting where some factors have too many levels even for cell-level treatment—the worker and firm effects of Abowd et al. (1999), hospitals, aircraft, municipality–year interactions—and are removed by *absorption* rather than estimation. The engine composes with the standard absorption device: augment the linear predictor with G absorbed factors $\alpha^{(g)}_{g(i)}$, concentrate them out by cycling closed-form profile updates (for Poisson, $\alpha^{(g)}_a = \log\{\sum_{i:g(i)=a} y_i / \sum_{i:g(i)=a} \exp(\eta_i - \alpha^{(g)}_a)\}$, the multiplicative update of Guimarães and Portugal, 2010; Bergé, 2018; Correia et al., 2020, used at scale in three-way gravity estimation by Larch et al., 2019), and take each tilted-moment Newton step on $(\boldsymbol{\gamma}, \boldsymbol{\delta})$ at the profiled effects, with the inner cycle re-solved at every trial point of the line search (Algorithm S1 of the supplement). Each inner sweep is $O(N)$; with a single absorbed factor the inner solve is exact in one sweep. The composition inherits the engine's economics: absorbed factors, like cells, never enter any design matrix.

This inner–outer scheme is the algorithm used across the HDFE–GLM software ecosystem, and its convergence status deserves a precise statement. Convergence of the *inner* loop—iterative demeaning over the absorbed factors at fixed weights—is classical: it is the method of alternating projections, established for two subspaces by von Neumann (1950) and for arbitrarily many by Halperin (1962), with the Kaczmarz sweep (Kaczmarz, 1937) as the coordinate special case (see Escalante and Raydan, 2011 for a monograph treatment and Deutsch and Hundal, 1997 for rates), and it was brought into fixed-effects estimation on exactly this foundation by Gaure (2013). The *combined* nonlinear cycle—profile updates of the absorbed effects alternated with Newton/scoring steps on a GLM likelihood—has, by contrast, been validated by simulation and practice (Guimarães and Portugal, 2010; Bergé, 2018; Correia et al., 2020) rather than by proof. The following theorem supplies the missing unified statement for the combined inner–outer iteration; its proof, and the finer rate and asymptotic theory, are in the online supplement.

Theorem 2 (Global convergence of the absorbed-factor iteration)**.** *Suppose the link is canonical, or more generally log-concave in the sense that each per-observation contribution $\eta \mapsto \ell_i(\eta)$ is strictly concave; suppose the maximum-likelihood estimate exists (no separation, which the drop rule below enforces for the absorbed blocks); and suppose the non-absorbed information block is nonsingular on the initial superlevel set. Then the iterates of the inner–outer scheme above—closed-form or exact one-dimensional block maximizations of the absorbed effects, alternated with Armijo-damped tilted-moment Newton/scoring steps for $(\boldsymbol{\gamma}, \boldsymbol{\delta})$—increase the likelihood monotonically and converge to the maximum-likelihood estimate, from any starting value, for any (essentially) cyclic ordering of the blocks.*

The proof (Supplement Theorem S1) is a block-coordinate-ascent argument on the concave profile objective, handling the parameter space's flat directions and the inexact (damped-Newton) outer block; the log-concave-link generality covers gamma–log and probit as well as the canonical families. The supplement further gives the *local rate* (Theorems S2–S3): for two absorbed factors the per-sweep contraction equals $\cos^2$ of the Friedrichs angle between the weighted factor subspaces—the squared second singular value of a normalized weighted cross-tabulation—and for $G \geq 3$ it is the spectral radius of a level-space sweep operator built from the pairwise cross-tabulations alone, computable by power iteration before touching the microdata, with an angle-based product certificate; observed contraction factors match these predictions to three decimals. Because the estimator *is* the MLE, its asymptotics are the MLE's: classical at fixed dimension, and, in the proportional regime of one Poisson fixed effect per bounded-size cell ($J/N \to c \in (0,1)$), unbiased with valid cell-clustered standard errors, because profiling then coincides with conditioning (Theorems S4–S6; cf. the incidental-parameter analysis of Fernández-Val and Weidner, 2016). Separated absorbed groups (no events) are detected and dropped a priori—the level-wise instance of the drop rule of Correia et al. (2019), whose linear-programming certificate treats separation in full generality and is implemented in `ppmlhdfe` (Correia et al., 2020); Section 6.2 exercises the full composition at scale, and Section 6.1 checks it against `pyfixest` on real data.

5 Numerical experiments

All experiments run on a single laptop-class core of the Python/NumPy reference implementation and complete in minutes; every number below regenerates from a fixed seed (scripts in the code supplement). The studies are organized around three comparisons: dense Newton as the ground-truth baseline for exactness (Section 5.1); sparse-design IRLS as the strongest general-purpose competitor, together with the production solvers `statsmodels` and `glum` (Section 5.2) and, for the absorbed path, `pyfixest` (Section 5.3); and the scaling, stability, and sensitivity envelope (Sections 5.4–5.6). The studies validating the supplement's absorption, asymptotic, and separation theory are reported in Section S4 of the online supplement.

5.1 Exactness and per-iteration cost

Study 1 fits Poisson, logistic, and gamma models with $J \in \{50, 200, 500\}$ cells, $p \in \{2, 5\}$ continuous covariates, and N up to 10^6, tracing the full iterate sequence of tilted-moment Newton and of textbook dense Newton from the same starting point. Table 1 reports the maximum deviation between the two iterate sequences across all iterations, the final-estimate deviation, and the per-iteration wall-clock times.

Three observations. First, the iterate sequences agree to 10^{-13}–10^{-15} throughout—floating-point reordering noise—confirming Theorem 1(iv): the tilted iteration is the same algorithm as full-design Newton, not an approximation to it. Second, measured per-iteration speedups range

Table 1: Study 1: tilted-moment versus dense Newton. "Iterate dev." is the maximum absolute difference between the two iterate sequences over all iterations and coordinates; times are medians per Newton iteration. Selected configurations; the full grid is in the supplement.

Family	N	J	p	Iterate dev.	Final dev.	t_{dense} (s)	t_{tilted} (s)
Poisson	2×10^5	50	2	5.1×10^{-14}	2.4×10^{-15}	0.119	0.010
Poisson	2×10^5	200	2	5.1×10^{-14}	3.7×10^{-15}	0.503	0.010
Poisson	2×10^5	500	2	6.0×10^{-13}	3.0×10^{-14}	1.481	0.017
Poisson	10^6	200	2	7.4×10^{-14}	1.1×10^{-14}	2.490	0.062
Poisson	2×10^5	200	5	1.1×10^{-13}	1.9×10^{-15}	0.520	0.016
Logistic	2×10^5	500	2	2.2×10^{-14}	4.0×10^{-15}	1.738	0.015
Logistic	10^6	200	2	8.2×10^{-15}	2.2×10^{-16}	2.918	0.082
Gamma	2×10^5	500	2	1.3×10^{-13}	5.2×10^{-15}	1.443	0.013
Gamma	10^6	200	2	2.5×10^{-13}	1.3×10^{-15}	2.662	0.058

from 12× ($J = 50$) to 117× ($J = 500$), growing with k as Corollary 1 predicts, though below the raw FLOP ratios (430×–4700×) because the dense baseline runs BLAS-3 while the tilted pass is memory-bound—the FLOP ratio is the right guide to scaling, the measured ratio to wall-clock budgeting. Third, the dense baseline at $N = 10^6$, $J = 200$ already needs a 1.6 GB design matrix where the tilted pass needs the p continuous columns only; at the dimensions of Section 6.2 the dense fit exceeds the memory of a typical workstation before the first iteration.

5.2 Comparison with production implementations

Study 1b benchmarks total time and peak memory to convergence (identical tolerance 10^{-9}, comparable starting values) against the implementations a practitioner would actually reach for. Each fit runs in a fresh process so that peak resident set size (RSS) isolates one method's footprint; times are medians over repeated fits and iteration counts are reported alongside. For the non-absorbed model (Table 2, $N = 200{,}000$, $J = 200$, $p = 2$) the competitors are our own dense Newton; `statsmodels` (Seabold and Perktold, 2010), the standard general-purpose Python GLM (dense IRLS); a sparse-design IRLS holding the design (continuous columns plus cell dummies) in compressed-sparse-row form and forming and solving the normal equations sparsely—the "the dummies are sparse, so use a sparse solver" route a careful analyst would code; and `glum` (Balzer et al., 2021), QuantCo's high-performance GLM, which carries categorical covariates in a dedicated split-matrix backend and is among the fastest Python GLM implementations.

Every method returns the same coefficients to 10^{-15}: the comparison is purely one of arithmetic organization and memory. The tilted pass is fastest and, more decisively, lightest: its ∼80 MB is the loaded columns plus cell accumulators, against ∼1 GB for a dense design and ∼4 GB for `statsmodels`' internal copies, a 12–50× memory advantage that grows with N (Section 5.4). The sparse-design IRLS is the strongest competitor at $p = 2$, as it should be—cell

Table 2: Study 1b, no absorbed factor: total time to convergence, peak resident memory, and iterations, with the maximum absolute coefficient deviation from the tilted fit. $N = 200{,}000$, $J = 200$ cells, $p = 2$, tolerance 10^{-9}; every method computes the same MLE. (t_{tilted} shown is the warm value; the very first subprocess spawned in a run pays a one-time interpreter/BLAS start-up cost.)

Family	Method	Time (s)	Peak (MB)	Iters	Max. dev.
Poisson	**tilted**	**0.08**	**80**	9	—
	sparse IRLS	0.26	107	9	6×10^{-17}
	`glum`	0.90	227	7	3×10^{-17}
	dense Newton	3.1	998	8	6×10^{-17}
	`statsmodels`	14.2	3972	7	2×10^{-15}
Logistic	**tilted**	**0.10**	**76**	8	—
	sparse IRLS	0.23	107	8	3×10^{-17}
	`glum`	0.37	229	5	2×10^{-16}
	dense Newton	3.1	997	8	6×10^{-17}
	`statsmodels`	14.0	3972	7	6×10^{-15}

Table 3: Study 1b, continuous-dimension sweep (Poisson, $N = 200{,}000$, $J = 200$): time and peak memory for the exact-MLE methods. The tilted cost grows in p on the continuous block only; the sparse Gram fill-in makes the sparse route super-linear in p.

	Time (s)			Peak (MB)		
p	tilted	dense	sparse	tilted	dense	sparse
2	0.08	3.0	0.27	80	998	107
20	0.41	3.4	2.73	135	1081	262
50	1.18	4.4	12.5	227	1219	512

dummies are the textbook sparse design—but the second panel of the study, which sweeps the continuous dimension $p \in \{2, 20, 50\}$ (Poisson; Table 3), shows where the two part ways: the tilted cost grows as $O(Np^2)$ on the dense continuous block only, while the sparse route's Gram matrix fills in and its factor grows super-linearly, so at $p = 50$ the sparse solver is 10× slower than tilted and dense Newton overtakes it. `statsmodels` and `glum` are omitted from the p-sweep: the former needs tens of seconds and gigabytes per fit, and `glum`'s unpenalized ($\alpha = 0$) solver returned a singular-matrix error for $p \geq 20$—it is engineered for penalized fitting, and the exact MLE is outside its robust regime.

5.3 High-dimensional fixed effects: comparison with `pyfixest`

When a high-dimensional factor is absorbed rather than entered as dummies, the relevant competitor is a dedicated HDFE–GLM tool. Study 1d (Table 4) benchmarks the absorbed-factor path against `pyfixest` (Bergé, 2018), the Python implementation of the `fixest` alternating-projections algorithm, on Poisson models with $N \in \{2, 5\} \times 10^5$, $J = 100$ cells, $p = 2$, and one

Table 4: Study 1d, one absorbed factor: time, peak memory, and deviation from the tilted fit. Dense-dummy Newton and `glum` form the $N \times (J + \text{levels})$ design explicitly and fail (singular Hessian, or out of memory / did not finish within 20 minutes) once the factor exceeds a few thousand levels. `pyfixest` matches to its own demeaning tolerance ($\sim 10^{-4}$).

Levels	Method	Time (s)	Peak (MB)	Max. dev.
500	**`tilted`**	**0.10**	**79**	—
	`pyfixest`	0.19	378	4×10^{-4}
	dense Newton	10.0	2837	4×10^{-13}
	`glum`	singular		—
5,000	**`tilted`**	**0.16**	**79**	—
	`pyfixest`	0.21	372	5×10^{-4}
	dense Newton	out of memory		—
	`glum`	singular		—
20,000	**`tilted`**	**0.36**	**112**	—
	`pyfixest`	0.64	544	4×10^{-4}
	dense Newton	out of memory		—
	`glum`	out of memory		—

absorbed factor of 500 to 20,000 levels; dense-dummy Newton and `glum` with dense dummies are included as references.

Against the specialized tool the tilted absorption is roughly twice as fast and four to five times lighter, while returning the exact MLE (`pyfixest` agrees to its demeaning tolerance). The dense-dummy routes confirm why HDFE tools exist at all: they exhaust memory once the factor passes a few thousand levels. The like-for-like comparison on *real* data in Section 6.1, where the two agree to 10^{-9}, is the more stringent test; the point of the synthetic sweep is the memory and timing trend.

5.4 Scaling in N and memory footprint

Study 1e runs the tilted Poisson fit at N from 10^6 to 5×10^7 ($J = 500$, $p = 2$), each in a fresh process (Table 5). Wall-clock is linear in N at $\approx$1.2 seconds per million records once the fixed start-up cost is amortized, and peak memory is linear as well—the loaded columns $O(N(p+1))$ plus the $O(J(k^2 + kp))$ accumulators—reaching 3.8 GB at $N = 5 \times 10^7$. A dense $N \times (p + J)$ design for the same problem would occupy 200 GB, a factor of 53 more; the tilted pass forms no design matrix, so the whole fit stays within a laptop's memory at a scale where the dense and sparse routes cannot allocate their first iterate. This is the controlled counterpart of the register-scale application (Section 6.2): the same $O(N)$ time and memory, measured across a two-order-of-magnitude range of N.

Table 5: Study 1e: tilted Poisson fit scaling ($J = 500$, $p = 2$). Time is linear in N; peak memory is linear and far below the size of a dense design, which for $N = 5 \times 10^7$ would need 200 GB.

N	Time (s)	Peak (GB)	Dense design (GB)	Ratio
1×10^6	1.4	0.17	4.0	24
5×10^6	6.9	0.44	20.1	46
2×10^7	23.6	1.56	80.3	51
5×10^7	59.6	3.79	200.8	53

5.5 Sensitivity: imbalance and rare events

Study 1f re-runs the exactness comparison under stress: cell sizes drawn from a Zipf(1.6) law (a few giant cells, a long tail of tiny ones) and event rates pushed to $\sim 0.2\%$ (Poisson means ~ 0.002; logistic prevalence $\sim 0.5\%$)—the regime where approximate methods resort to case-control subsampling (Fithian and Hastie, 2014)—with event-free cells dropped as the MLE requires. Across all eight family $\times$ scenario combinations the maximum deviation from dense Newton stays at 10^{-15}–10^{-14}, convergence takes at most 12 iterations (rare-event fits need a few more steps, as their likelihoods are flatter), and the streaming pass time is unchanged: the accumulation is a data-independent $O(N)$ sweep, so imbalance redistributes work across cells without changing its total. Exactness is an algebraic identity, so it is expected to survive stress; the study confirms that the *implementation*—binned accumulation, line search, drop rules—inherits it.

5.6 Numerical stability

Because the tilted iteration is the same Newton/Fisher iteration as the full-design fit (Theorem 1), the question a numerical analyst asks is whether the *implementation*—binned accumulation, log-domain likelihood, line search, pseudoinverse fallback—inherits the conditioning of the underlying problem or degrades it. Study 1g (Table 6) answers this on a battery of adversarial scenarios at $N = 10^5$: near-collinear covariates (correlation 0.9999); an ill-scaled covariate (multiplied by 10^6, giving information-matrix condition number $\sim 10^{17}$); extreme offsets spanning 30 log units; linear predictors driven to ± 40 (over/underflow territory); Zipf(1.3)-unbalanced cells; and an informative covariate carried on a 10^8 pedestal (catastrophic cancellation in any naive sum of squares). For each we report the condition number, the deviation from dense Newton on the same data, and the deviation from a cancellation-free reference—the mathematically identical model with the continuous columns standardized, then mapped back—so that a genuine conditioning failure is separated from a faithful reproduction of a hard problem.

The reading is clean. At condition numbers up to 10^{17}, with linear predictors at ± 40, exposures spanning 30 log units, near-collinear covariates, and Zipf-unbalanced cells, the tilted

Table 6: Study 1g: stability battery (Poisson; the logistic panel is similar). "κ" is the information-matrix condition number; "vs. dense" the max coefficient deviation from dense Newton on the raw data; "vs. std" the deviation from the standardized-covariate reference (a conditioning diagnostic). The tilted fit matches dense and the well-conditioned reference throughout, except under a 10^8 pedestal, where dense diverges *identically* (catastrophic cancellation shared by every double-precision solver) and centering the covariate—the one-line standardization the paper recommends—restores full accuracy.

Scenario	κ	vs. dense	vs. std	converged
balanced	4×10^5	2×10^{-14}	6×10^{-17}	yes
collinear (0.9999)	5×10^5	8×10^{-14}	1×10^{-13}	yes
ill-scaled ($\times 10^6$)	3×10^{17}	4×10^{-14}	0	yes
extreme offset	3×10^5	1×10^{-15}	6×10^{-17}	yes
huge η (± 40)	2×10^{11}	1×10^{-11}	3×10^{-17}	yes
unbalanced (Zipf)	3×10^5	1×10^{-15}	6×10^{-17}	yes
10^8 pedestal (raw)	9×10^{29}	1×10^7	0.27	no
10^8 pedestal (centered)	3×10^5	5×10^{-14}	3×10^{-17}	yes

fit reproduces dense Newton and the standardized reference to 10^{-11}–10^{-17}: the accumulation neither introduces nor amplifies ill-conditioning. The one genuine failure—an informative signal on a 10^8 pedestal—is a catastrophic-cancellation hazard of the raw parametrization, not of the tilted pass: dense Newton diverges by the same 10^7, and standardizing the covariate (which the reference implementation offers and the paper recommends as routine preprocessing, following standard practice; Higham, 2002) returns the fit to full accuracy. We flag this explicitly rather than suppress it: no double-precision GLM solver survives a covariate whose signal sits 16 digits below its magnitude, and the honest remedy is to standardize, not to claim immunity.

6 Case studies

6.1 On-time performance of NYC flights

The first application is to public data, chosen so that every number can be reproduced from a fresh download: the `nycflights13` data set (Wickham, 2021)—all 336,776 flights departing New York City's three airports in 2013, from the U.S. Bureau of Transportation Statistics. We model the probability of a late arrival (arrival delay $\geq$ 15 minutes, the FAA on-time criterion; 24.5% of the 327,346 flights with an observed delay) by logistic regression with a cell structure of carrier $\times$ origin $\times$ destination $\times$ month—J = 3,863 observed combinations, entering through a main-effects cell-level design with $k = 131$ parameters (16 carriers, 3 origins, 103 destinations, 12 months)—and $p = 2$ continuous covariates: scheduled departure time (in hours, centered) and log distance (standardized). One destination (LEX, a single on-time flight) is quasi-separated—its dummy's MLE is $-\infty$—and is dropped by the level-wise separation rule, after which the MLE exists.

The tilted logistic fit converges in 7 Newton iterations, 1.3 seconds including robust standard errors. The diurnal effect is the substantive headline: the odds of a late arrival multiply by 1.11 per hour of scheduled departure time (robust SE on the log scale 0.001)—delays cascade through the day, a finding familiar from the air-transportation literature, here estimated jointly with a full operational cell structure. Log distance illustrates the boundary case flagged in Section 3: distance is constant within origin–destination routes, hence within cells, so after destination main effects it is identified only through cross-route contrasts and is correspondingly imprecise (+0.65, robust SE 0.28)—the estimator handles cell-constant continuous covariates correctly, and the standard error reports how little such a covariate adds.

As external validation, the identical model fitted with `statsmodels` GLM on the dense 327,345 × 133 design (0.35 GB) gives a maximum absolute coefficient deviation of 2.3×10^{-13}, at 19 seconds versus 1.3: the same estimator, 15× cheaper, with no design matrix in memory.

A second model on the same data exercises the high-dimensional absorption path against `pyfixest`. We now absorb *aircraft*—the 3,692 tail numbers that survive the separation drop—crossed with a twelve-level month cell structure (an aircraft flies across months, so the two factors are identified), keeping the two continuous covariates. The tilted logistic fit with absorbed aircraft converges in 31 iterations and 1.5 seconds; `pyfixest`'s `feglm` on the same specification takes 2.1 seconds, and the two agree on the continuous coefficients to 1.2×10^{-9} (departure hour 0.103994 from both). This is the paper's most stringent external check: on real high-dimensional data, the tilted absorption reproduces the estimate of a mature, purpose-built HDFE–GLM package to nine digits, at comparable speed, while also carrying the cell-structured part of the model that `pyfixest` treats as ordinary covariates (script `app_flights.py`; the data file is fetched from the public repository cited in the script).

6.2 A register-scale stress test

The second case study is a full-composition benchmark at register scale, on synthetic data with a known generating process so that recovery can be verified exactly. The rate structure is calibrated to realistic all-cause mortality gradients (a Gompertz age slope, sex and education log-rate contrasts of 0.1–0.45, a secular trend), the generator is 40 lines of the code supplement, and every number below regenerates end to end in under a minute. The configuration exercises every element of Sections 2–4 simultaneously: absorbed factors, cell-level design, continuous covariates, separation handling, and cluster-robust inference.

The data comprise $N = 5{,}000{,}000$ person-year episodes (35,526 deaths; crude rate 9.5 per 1,000 person-years) with: municipality × year fixed effects ($500 \times 20 = 10{,}000$ groups) absorbed by the profile Gauss–Seidel scheme of Section 4.5 (Algorithm S1); a cell-level design of education (4) × sex (2) × age band (12), $J = 96$ cells, $k = 16$ parameters; and $p = 2$ continuous covariates—standardized log income and a standardized pollution-exposure index correlated with the municipality effects (making the fixed effects non-ignorable for it). Standard errors are clustered on the 500 municipalities.

The fit: 322 zero-event municipality–year groups (159,464 episodes) are dropped by the separation rule of Section 4.5; the remaining model—9,678 absorbed effects, 16 cell-level and 2 individual-level coefficients—converges in 8 tilted Newton iterations, 11.2 seconds end to end on one core. A dense dummy-variable design for the same model would occupy $N \times 9{,}696$ doubles $\approx 0.4\,\text{TB}$ before the first iteration. For verification at feasible scale, the subpopulation of the first 10 municipalities ($n = 97{,}723$, 196 absorbed groups) was fitted both ways: maximum absolute coefficient deviation 4.0×10^{-11}, tilted 0.1s versus dense 3.0s.

Substantively, the model recovers the calibrated truth within sampling error: income -0.180 (cluster SE 0.005; truth -0.18), pollution $+0.049$ (SE 0.005; truth $+0.06$, within two SEs, identified from within-municipality variation only, as it must be after absorption), female -0.455 (truth -0.45), education gradients $-0.13/-0.27/-0.40$ (truth $-0.12/-0.25/-0.40$), and a monotone age gradient tracking the Gompertz truth to two decimals. The point of the exercise is not these numbers—the generator made them true—but the computational demonstration: a model of this shape, with inference, is an interactive-latency computation under the tilted-moment organization, whereas a dense dummy-variable design for the same model would need to allocate a terabyte-scale matrix before its first iteration.

7 Discussion

The paper establishes a single algorithmic result and measures its consequences. For any exponential dispersion family whose design mixes cell-constant categorical structure with continuous covariates, the score vector and information matrix at an arbitrary parameter value—and with them exact Newton and Fisher-scoring steps, profile Hessians, and sandwich and cluster-robust covariance matrices—are linear combinations of three tilted within-cell moments that one streaming $O(N(p^2 + p + 1))$ pass produces (Theorem 1, Corollary 1). The likelihood *function* admits no such reduction (Proposition 1); the iteration does, and the gap between those two statements is the method. The experiments bound its value against each relevant alternative: exactness against dense Newton at 10^{-13}–10^{-15}; cost and memory against the strongest general-purpose competitor (sparse-design IRLS) and against production solvers; $O(N)$ scaling to 5×10^7 observations; and accuracy across a stability battery reaching condition numbers of 10^{17}. Because the accumulators are additive across data partitions, the pass extends directly to distributed and out-of-core execution; a tuned (threaded, cache-blocked, or GPU segmented-reduction) implementation of the accumulation kernel is the natural engineering continuation.

High-dimensional fixed-effect absorption served as the stress test, and the division of labor bears repeating: absorbed factors handle dummy structure too large even for cells, the tilted pass handles everything else, and neither ever enters a design matrix. The inner demeaning loop's convergence is classical alternating-projections theory (von Neumann, 1950; Halperin, 1962; Gaure, 2013); the contribution here is the unified guarantee for the combined nonlinear cycle (Theorem 2) and, in the supplement, its local rates from pairwise cross-tabulations, asymptotics up to the proportional regime $J/N \to c$, and Firth and penalized variants computed inside the same pass. These results are kept out of the main text so that the tilting engine stands as the paper's subject.

Software. All results regenerate from a self-contained open-source Python/NumPy reference implementation (`tilted_glm.py`, roughly 800 lines) released with the manuscript and organized so that the correspondence between code and stated results is line-by-line auditable; the benchmark harness fits every competitor in an isolated subprocess with peak-memory accounting. A companion Stata command (`stpois`), distributed in the same repository, implements the Poisson event-history case for applied use. Both are available at `https://github.com/andreasljungstrm/stpois`.

Acknowledgments. The author thanks colleagues at SOFI for helpful discussions.

A Proofs for Section 2

A.1 Proof of Proposition 1

(i) If $(\mathbf{x}_i, o_i) = (\bar{\mathbf{x}}_j, \bar{o}_j)$ within cells, then $\eta_i = \boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j + \bar{o}_j$ is constant within cells and (3) becomes $\sum_j \{(\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j + \bar{o}_j)D_j - n_j\, b(\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j + \bar{o}_j)\}$ up to constants, a function of the displayed cell statistics for any cumulant b; the collapsed likelihood is the microdata likelihood up to an additive constant, so their maximizers and Hessians coincide. For the Poisson family the exposure offset need not be constant, since $\sum_{i\in j} e^{\eta_i} = T_j\, e^{\boldsymbol{\gamma}'\bar{\mathbf{x}}_j + \boldsymbol{\delta}'\mathbf{w}_j}$ collects the offsets into T_j.

(ii) Writing $s_j = \boldsymbol{\delta}'\mathbf{w}_j$, the log likelihood is, up to constants,

$$\ell(\boldsymbol{\gamma}, \boldsymbol{\delta}) = \sum_j \Big\{\boldsymbol{\gamma}' S_j + s_j D_j + \textstyle\sum_{i\in j} o_i y_i - G_j(s_j, \boldsymbol{\gamma})\Big\}, \qquad G_j(s, \boldsymbol{\gamma}) = \textstyle\sum_{i\in j} b(s + \boldsymbol{\gamma}'\mathbf{x}_i + o_i),$$

which establishes computability from $\{S_j, D_j, G_j\}$. Conversely, suppose a statistic permits evaluation of ℓ on an open set $\mathcal{O} \ni (\boldsymbol{\gamma}_0, \boldsymbol{\delta}_0)$. Provided the map $j \mapsto \mathbf{w}_j$ is injective (cells are distinct covariate combinations) and $\boldsymbol{\delta} \mapsto (s_1, \ldots, s_J)$ has open range on $\mathcal{O}$'s $\boldsymbol{\delta}$-slice—as holds whenever the $\mathbf{w}_j$ are affinely independent, and in general after passing to the identified subfamily—one recovers each $G_j(\cdot, \boldsymbol{\gamma})$ on an open interval of s for each $\boldsymbol{\gamma}$ in the slice. Each $G_j(\cdot, \boldsymbol{\gamma})$ is a finite sum of translates of the real-analytic b, hence real-analytic on $\mathbb{R}$; knowledge on an open interval therefore determines it on the whole line by analytic continuation. The boundary hypothesis $b(u) = e^u(1 + o(1))$ as $u \to -\infty$ then gives

$$e^{-s}\, G_j(s, \boldsymbol{\gamma}) = \sum_{i \in j} e^{-s}\, b(s + \boldsymbol{\gamma}'\mathbf{x}_i + o_i) \;\longrightarrow\; \sum_{i \in j} e^{\boldsymbol{\gamma}'\mathbf{x}_i + o_i} = \mathcal{L}_{\nu_j}(\boldsymbol{\gamma}), \qquad s \to -\infty,$$

termwise (for Poisson the relation is exact at every s; for Bernoulli it is the expansion $\log(1+\varepsilon) = \varepsilon(1 + o(1))$). Hence the statistic determines $\mathcal{L}_{\nu_j}$ on an open subset of $\mathbb{R}^p$; $\mathcal{L}_{\nu_j}$ is entire because ν_j is finite and discrete, so it is determined everywhere by analytic continuation, and the Laplace transform of a finite positive measure determines the measure. The statistic therefore determines ν_j, whose minimal description has dimension equal to (one plus p times) its support size.

(iii) Jensen's inequality applied to the strictly convex map $z \mapsto e^z$ under the probability measure $\nu_j/\nu_j(\mathbb{R}^p)$ gives $\nu_j(\mathbb{R}^p)^{-1}\mathcal{L}_{\nu_j}(\boldsymbol{\gamma}) \geq \exp(\boldsymbol{\gamma}'\bar{\mathbf{x}}_j)$ with equality iff $\boldsymbol{\gamma}'\mathbf{x}$ is ν_j-a.s. constant. □

A.2 Proof of Theorem 1

Differentiating (3) with $\theta_i = \eta_i$: $\phi\, \partial\ell/\partial\boldsymbol{\gamma} = \sum_i (y_i - b'(\eta_i))\mathbf{x}_i$ and $\phi\, \partial\ell/\partial\boldsymbol{\delta} = \sum_i (y_i - b'(\eta_i))\mathbf{w}_{j(i)}$. Grouping the second sum by cells and using that $\mathbf{w}_j$ is constant within cell j gives $\sum_j \mathbf{w}_j R_j$; grouping the first gives $\sum_i y_i\mathbf{x}_i - \sum_j \mathbf{m}_j^{b'}$ with $\mathbf{m}_j^{b'} = \sum_{i \in j} b'(\eta_i)\mathbf{x}_i = B_j \mathbb{E}_{P^{b'}_{j,\boldsymbol{\theta}}}[\mathbf{x}]$ by the definition of the b'-tilt. Differentiating again, $\phi\,\mathbf{F} = \sum_i b''(\eta_i)\,\mathbf{a}_i\mathbf{a}_i'$ with $\mathbf{a}_i = (\mathbf{x}_i', \mathbf{w}_{j(i)}')'$; expanding the blocks and grouping by cells yields (ii), and the moment identities follow from Definition 1. For (iii), $b' = b'' = \exp$ gives $\omega_i = e^{\boldsymbol{\delta}'\mathbf{w}_j}\, t_i e^{\boldsymbol{\gamma}'\mathbf{x}_i}$, the cell factor cancels in the normalization, and $M_j^{(0)} = e^{\boldsymbol{\delta}'\mathbf{w}_j}\mathcal{L}_{\nu_j}(\boldsymbol{\gamma})$; the cumulant identities $\mathbb{E}_{P_{j,\boldsymbol{\gamma}}}[\mathbf{x}] = \nabla K_j$ and $\mathrm{Var} = \nabla^2 K_j$ are the standard derivatives of a log-Laplace transform, and $\mathbb{E}[\mathbf{x}\mathbf{x}'] = \mathrm{Var} + \mathbb{E}\mathbb{E}'$ gives the stated $\boldsymbol{\gamma}$-block. (iv) The score and information assembled from (i)–(ii) are, entry for entry, the same matrices as those computed from the stacked design; hence so is the Newton step. □

A.3 Proof of Corollary 1

The microdata pass evaluates, per observation, η_i from the cached cell linear predictor plus $\boldsymbol{\gamma}'\mathbf{x}_i$ ($O(p)$), the scalars $b'(\eta_i), b''(\eta_i)$ ($O(1)$), the binned accumulations of ω_i and $\omega_i\mathbf{x}_i$ into cell $j(i)$ ($O(p)$), and the symmetric rank-one update $\omega_i\mathbf{x}_i\mathbf{x}_i'$ ($O(p^2)$ scalar multiply–adds, $p(p+1)/2$ distinct); total $O(N(p^2+p+1))$. Assembly of $\sum_j M_j^{(0)}\mathbf{w}_j\mathbf{w}_j'$ and $\sum_j \mathbf{m}_j\mathbf{w}_j'$ costs $O(J(k^2+kp))$, and the symmetric solve $O((p+k)^3)$. The dense iteration forms $\sum_i \omega_i\mathbf{a}_i\mathbf{a}_i'$ at $O(N(p+k)^2)$ plus the same solve. Ratios follow. With $p = 0$ the pass reduces to two binned sums (R_j, $M_j^{(0)}$), and if the offsets enter through cell sums (pure rate models), even these are fixed after one pass. □

A.4 Proof of Proposition 2

The ML score for a general link is $\sum_i (y_i - \mu_i) \frac{\partial \mu_i / \partial \eta_i}{V(\mu_i)} \mathbf{a}_i = \sum_i (y_i - \mu_i)\, g'(\mu_i)\, \omega_i\, \mathbf{a}_i = \sum_i r_i \mathbf{a}_i$, using $\partial \mu / \partial \eta = 1/g'(\mu)$, and the expected information is $\sum_i \omega_i \mathbf{a}_i \mathbf{a}_i'$ (McCullagh and Nelder, 1989, Sec. 2.5). These have the same cell-decomposable form as in Theorem 1 with (ω_i, r_i) in place of $(b''(\eta_i), y_i - b'(\eta_i))$, so the accumulation and complexity claims carry over. A fixed point of the scoring iteration has $\sum_i r_i \mathbf{a}_i = \mathbf{0}$, i.e., is a root of the ML score equations. For the gamma–log pair, $V(\mu) = \mu^2$ and $g'(\mu) = 1/\mu$ give $\omega_i = 1/(\mu_i^2 \mu_i^{-2}) = 1$; the moments (5) are then constant in $\boldsymbol{\theta}$, and only $R_j = \sum_{i \in j} (y_i - \mu_i)/\mu_i$ requires re-evaluation, an $O(Np)$ pass. $\square$

References

Abowd, J. M., F. Kramarz, and D. N. Margolis (1999). High wage workers and high wage firms. *Econometrica 67*(2), 251–333.

Aitkin, M. and D. Clayton (1980). The fitting of exponential, Weibull and extreme value distributions to complex censored survival data using GLIM. *Journal of the Royal Statistical Society: Series C 29*(2), 156–163.

Albert, A. and J. A. Anderson (1984). On the existence of maximum likelihood estimates in logistic regression models. *Biometrika 71*(1), 1–10.

Andersen, P. K., Ø. Borgan, R. D. Gill, and N. Keiding (1993). *Statistical Models Based on Counting Processes*. New York: Springer.

Balzer, J., M.-A. Orłowski, and QuantCo (2021). glum: High-performance generalized linear models in Python. *Open-source software, https://github.com/Quantco/glum*.

Barndorff-Nielsen, O. E. (1978). *Information and Exponential Families in Statistical Theory*. Chichester: Wiley.

Baydin, A. G., B. A. Pearlmutter, A. A. Radul, and J. M. Siskind (2018). Automatic differentiation in machine learning: A survey. *Journal of Machine Learning Research 18*(153), 1–43.

Bergé, L. (2018). Efficient estimation of maximum likelihood models with multiple fixed-effects: The R package FENmlm. CREA Discussion Paper 2018-13, CREA, University of Luxembourg.

Breslow, N. E. and N. E. Day (1987). *Statistical Methods in Cancer Research, Vol. II: The Design and Analysis of Cohort Studies*. Number 82 in IARC Scientific Publications. Lyon: International Agency for Research on Cancer.

Byrd, R. H., G. M. Chin, W. Neveitt, and J. Nocedal (2011). On the use of stochastic Hessian information in optimization methods for machine learning. *SIAM Journal on Optimization 21*(3), 977–995.

Cameron, A. C., J. B. Gelbach, and D. L. Miller (2011). Robust inference with multiway clustering. *Journal of Business & Economic Statistics 29*(2), 238–249.

Clayton, D. and M. Hills (1993). *Statistical Models in Epidemiology.* Oxford: Oxford University Press.

Correia, S. (2016). A feasible estimator for linear models with multi-way fixed effects. Working paper, Duke University.

Correia, S., P. Guimarães, and T. Zylkin (2019). Verifying the existence of maximum likelihood estimates for generalized linear models. *arXiv preprint arXiv:1903.01633*.

Correia, S., P. Guimarães, and T. Zylkin (2020). Fast Poisson estimation with high-dimensional fixed effects. *The Stata Journal 20*(1), 95–115.

Daniels, H. E. (1954). Saddlepoint approximations in statistics. *The Annals of Mathematical Statistics 25*(4), 631–650.

Deutsch, F. and H. Hundal (1997). The rate of convergence for the method of alternating projections, II. *Journal of Mathematical Analysis and Applications 205*(2), 381–405.

Drineas, P. and M. W. Mahoney (2016). RandNLA: Randomized numerical linear algebra. *Communications of the ACM 59*(6), 80–90.

Efron, B. (1981). Nonparametric standard errors and confidence intervals. *Canadian Journal of Statistics 9*(2), 139–158.

Escalante, R. and M. Raydan (2011). *Alternating Projection Methods.* Fundamentals of Algorithms. Philadelphia: SIAM.

Fernández-Val, I. and M. Weidner (2016). Individual and time effects in nonlinear panel models with large N, T. *Journal of Econometrics 192*(1), 291–312.

Firth, D. (1993). Bias reduction of maximum likelihood estimates. *Biometrika 80*(1), 27–38.

Fithian, W. and T. Hastie (2014). Local case-control sampling: Efficient subsampling in imbalanced data sets. *The Annals of Statistics 42*(5), 1693–1724.

Frome, E. L. (1983). The analysis of rates using Poisson regression models. *Biometrics 39*(3), 665–674.

Gaure, S. (2013). OLS with multiple high dimensional category variables. *Computational Statistics & Data Analysis 66*, 8–18.

Golub, G. H. and C. F. Van Loan (2013). *Matrix Computations* (4th ed.). Baltimore: Johns Hopkins University Press.

Gouriéroux, C., A. Monfort, and A. Trognon (1984). Pseudo maximum likelihood methods: Applications to Poisson models. *Econometrica 52*(3), 701–720.

Green, P. J. (1984). Iteratively reweighted least squares for maximum likelihood estimation, and some robust and resistant alternatives. *Journal of the Royal Statistical Society: Series B 46*(2), 149–192.

Guimarães, P. and P. Portugal (2010). A simple feasible procedure to fit models with high-dimensional fixed effects. *The Stata Journal 10*(4), 628–649.

Halperin, I. (1962). The product of projection operators. *Acta Scientiarum Mathematicarum 23*, 96–99.

Heinze, G. and M. Schemper (2002). A solution to the problem of separation in logistic regression. *Statistics in Medicine 21*(16), 2409–2419.

Higham, N. J. (2002). Accuracy and stability of numerical algorithms.

Holford, T. R. (1980). The analysis of rates and of survivorship using log-linear models. *Biometrics 36*(2), 299–305.

Huber, P. J. (1967). The behavior of maximum likelihood estimates under nonstandard conditions. In *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*, Volume 1, pp. 221–233. Berkeley: University of California Press.

Jaggi, M., V. Smith, M. Takáč, J. Terhorst, S. Krishnan, T. Hofmann, and M. I. Jordan (2014). Communication-efficient distributed dual coordinate ascent. *Advances in Neural Information Processing Systems (NeurIPS) 27*.

Jensen, J. L. (1995). *Saddlepoint Approximations*. Oxford: Oxford University Press.

Jørgensen, B. (1987). Exponential dispersion models (with discussion). *Journal of the Royal Statistical Society: Series B 49*(2), 127–162.

Kaczmarz, S. (1937). Angenäherte Auflösung von Systemen linearer Gleichungen. *Bulletin International de l'Académie Polonaise des Sciences et des Lettres 35*, 355–357.

Kitamura, Y. and M. Stutzer (1997). An information-theoretic alternative to generalized method of moments estimation. *Econometrica 65*(4), 861–874.

Kosmidis, I. and D. Firth (2009). Bias reduction in exponential family nonlinear models. *Biometrika 96*(4), 793–804.

Larch, M., J. Wanner, Y. V. Yotov, and T. Zylkin (2019). Currency unions and trade: A PPML re-assessment with high-dimensional fixed effects. *Oxford Bulletin of Economics and Statistics 81*(3), 487–510.

Liang, K.-Y. and S. L. Zeger (1986). Longitudinal data analysis using generalized linear models. *Biometrika 73*(1), 13–22.

Lumley, T. (2011). biglm: Bounded memory linear and generalized linear models. *R package version 0.9*.

Ma, P., M. W. Mahoney, and B. Yu (2015). A statistical perspective on algorithmic leveraging. *Journal of Machine Learning Research 16*, 861–919.

Martens, J. (2010). Deep learning via Hessian-free optimization. *Proceedings of the 27th International Conference on Machine Learning (ICML)*, 735–742.

McCullagh, P. and J. A. Nelder (1989). *Generalized Linear Models* (2nd ed.). London: Chapman & Hall.

Nelder, J. A. and R. W. M. Wedderburn (1972). Generalized linear models. *Journal of the Royal Statistical Society: Series A 135*(3), 370–384.

Nocedal, J. and S. J. Wright (2006). *Numerical Optimization* (2nd ed.). New York: Springer.

Owen, A. B. (2001). *Empirical Likelihood.* Boca Raton: Chapman & Hall/CRC.

Pearlmutter, B. A. (1994). Fast exact multiplication by the Hessian. *Neural Computation 6*(1), 147–160.

Santos Silva, J. M. C. and S. Tenreyro (2006). The log of gravity. *The Review of Economics and Statistics 88*(4), 641–658.

Santos Silva, J. M. C. and S. Tenreyro (2010). On the existence of the maximum likelihood estimates in Poisson regression. *Economics Letters 107*(2), 310–312.

Santos Silva, J. M. C. and S. Tenreyro (2011). poisson: Some convergence issues. *The Stata Journal 11*(2), 207–212.

Schennach, S. M. (2007). Point estimation with exponentially tilted empirical likelihood. *The Annals of Statistics 35*(2), 634–672.

Seabold, S. and J. Perktold (2010). statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*, 92–96.

Stammann, A. (2018). Fast and feasible estimation of generalized linear models with high-dimensional k-way fixed effects. *arXiv preprint arXiv:1707.01815*.

von Neumann, J. (1950). *Functional Operators, Vol. II: The Geometry of Orthogonal Spaces*. Number 22 in Annals of Mathematics Studies. Princeton: Princeton University Press. reprint of 1933 lecture notes; cited in the fixed-effects literature as von Neumann (1949).

Wang, H., M. Yang, and J. Stufken (2019). Information-based optimal subdata selection for big data linear regression. *Journal of the American Statistical Association 114*(525), 393–405.

Wang, H., R. Zhu, and P. Ma (2018). Optimal subsampling for large sample logistic regression. *Journal of the American Statistical Association 113*(522), 829–844.

Wedderburn, R. W. M. (1974). Quasi-likelihood functions, generalized linear models, and the Gauss–Newton method. *Biometrika 61*(3), 439–447.

Wedderburn, R. W. M. (1976). On the existence and uniqueness of the maximum likelihood estimates for certain generalized linear models. *Biometrika 63*(1), 27–32.

Welford, B. P. (1962). Note on a method for calculating corrected sums of squares and products. *Technometrics 4*(3), 419–420.

White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica 48*(4), 817–838.

White, H. (1982). Maximum likelihood estimation of misspecified models. *Econometrica 50*(1), 1–25.

Wickham, H. (2021). *nycflights13: Flights that Departed NYC in 2013*. R package version 1.0.2; data from the U.S. Bureau of Transportation Statistics.

Zhang, Y., J. C. Duchi, and M. J. Wainwright (2013). Communication-efficient algorithms for statistical optimization. *Journal of Machine Learning Research 14*, 3321–3363.